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## AI on the Factory Floor: Evidence from Manufacturing Job Postings

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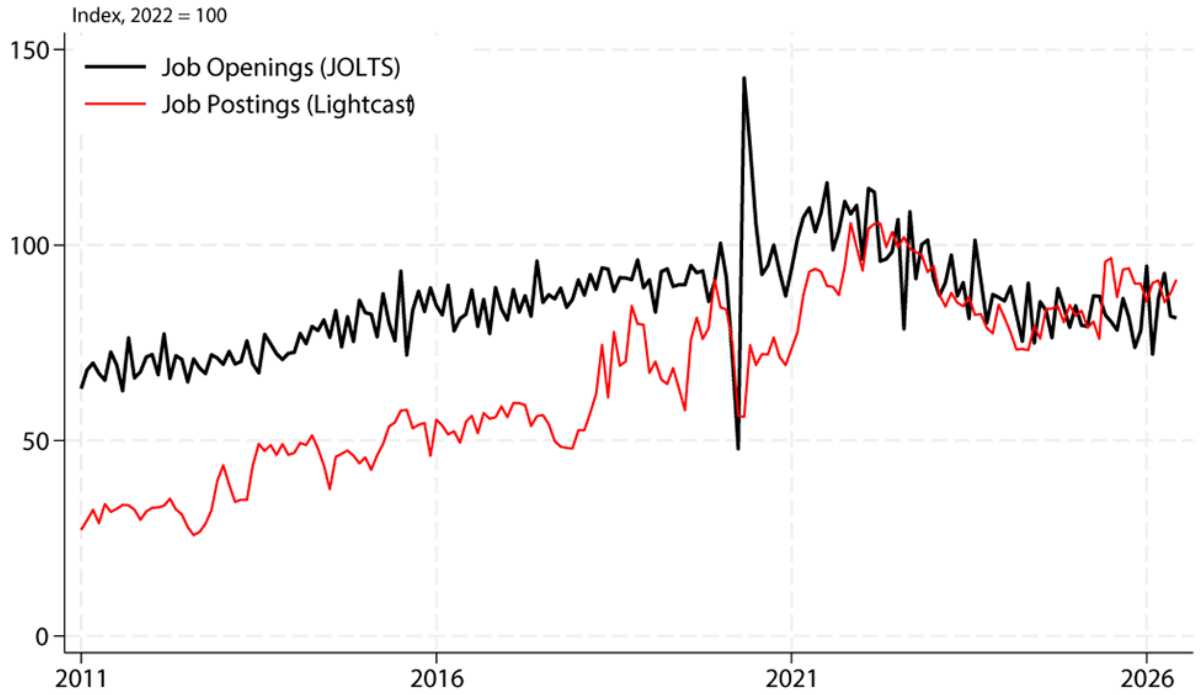
Artificial intelligence (AI) technologies have the potential to transform production processes and labor markets, yet their ultimate effects on employment, occupations, and skill demand remain uncertain (Acemoglu and Restrepo, 2018). While research continues to explore implications of AI adoption for productivity and employment, firms' hiring decisions provide a timely measure of employers' evolving skill requirements (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Acemoglu et al., 2022). This note examines whether and how AI adoption is affecting skill requirements by analyzing detailed job posting data from the manufacturing sector.

Several features make manufacturing particularly relevant for this analysis. Most task-based measures of AI exposure classify manufacturing as having relatively lower exposure than many other sectors since production work relies heavily on physical tasks rather than cognitive activities (Felten, Raj, and Seamans, 2021; Eloundou et al., 2023). Yet manufacturers have historically been early adopters of new production technologies. Recent evidence from the U.S. Census Bureau's Business Trends and Outlook Survey (BTOS) shows that a growing share of manufacturing establishments report using AI in their operations, in line with the national average, though adoption rates remain below sectors like Information and Finance. These developments raise an important question: does the growing adoption of AI translate into changing skill requirements when manufacturers hire workers?

This note examines this question using Lightcast data on online job postings for manufacturing establishments. Lightcast aggregates online job postings from company career pages, job boards, and recruitment websites, providing posting-level detail on occupation, industry, employer, location, advertised skills, and wages (when reported). Unlike the Bureau of Labor Statistics' (BLS) Job Openings and Labor Turnover Survey (JOLTS), which measures employer-reported vacant positions, Lightcast captures employers' recruiting activity through online job advertisements. Consequently, job postings should be interpreted as a timely indicator of hiring demand rather than a direct measure of job openings.

With these caveats in mind, figure 1 compares aggregate Lightcast job postings with job openings reported in JOLTS. Although the two series measure different concepts and Lightcast's coverage expanded over time, particularly as online recruiting has become more prevalent, they have been moving closely together in recent years.

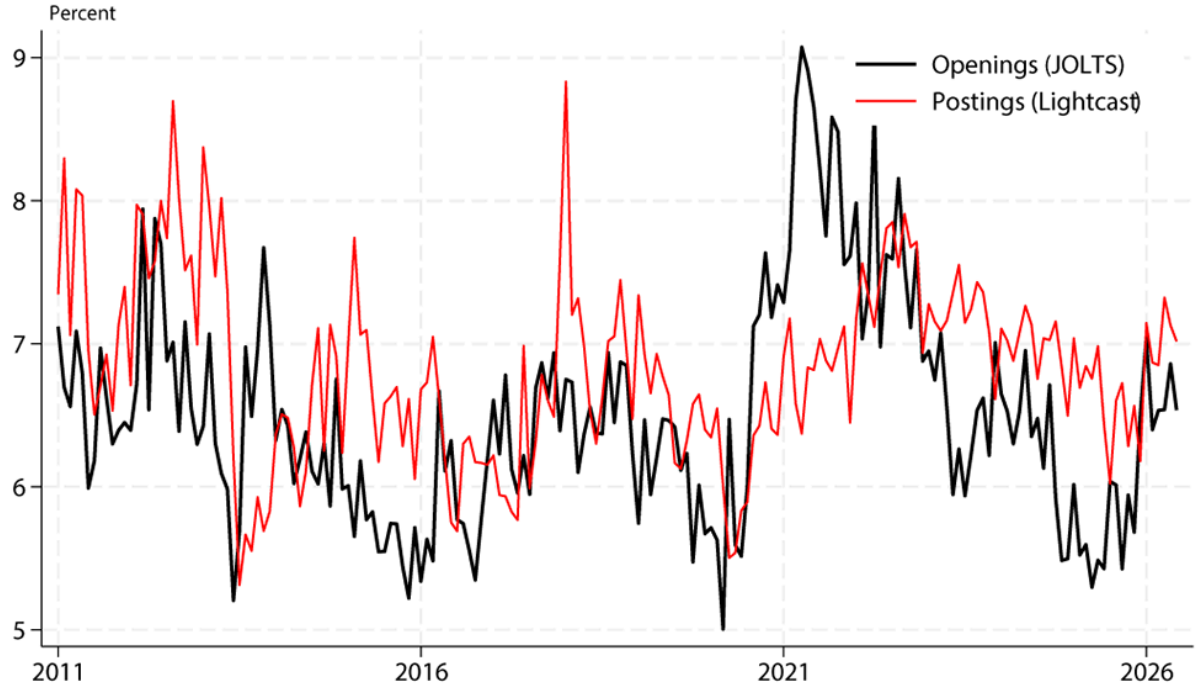
Figure 1. New Job Openings and New Job Postings




Note: New job openings are estimated using the model from Davis, Faberman, and Haltiwanger (2013). Data through June 2026.
Source: Bureau of Labor Statistics (BLS) via HaverAnalytics; Lightcast, Job Vacancy Data; authors' calculations.
Accessible version

Because the analysis focuses on manufacturing, figure 2 compares the manufacturing share of new job postings in Lightcast (red line) with the manufacturing share of job openings from JOLTS (black line). Manufacturing accounts for approximately 6-1/2 percent of new hiring demand in both datasets, and the two series closely track one another over time.[2] These findings suggest that Lightcast captures manufacturing hiring activity comparably to JOLTS while providing substantially richer information on occupations, skills, firms, and advertised wages.

Figure 2. Manufacturing Share of New Job Openings and New Job Postings




Note: Share of total openings and of new postings for the manufacturing sector relative to the universe of openings and postings. Data through June 2026.
Source: BLS via HaverAnalytics; Lightcast, Job Vacancy Data.
Accessible version

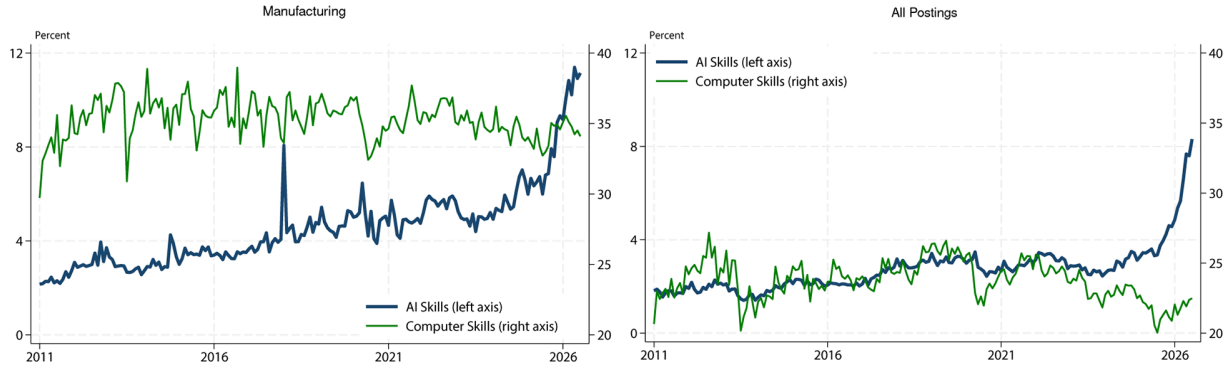
The remainder of the note uses the detailed Lightcast posting-level data to document how manufacturers' demand for AI-related and broader computer skills has evolved across manufacturing occupations and how those skills are reflected in advertised wages. By examining the skills employers seek rather than the tasks workers currently perform, the analysis complements task-based measures of AI exposure and provides new evidence on how manufacturers are adapting their hiring practices as AI technologies diffuse.

Within this framework, computer skills provide a useful benchmark because they capture the skill requirements associated with a mature, widely adopted technology, whereas AI-related skills reflect the emergence of a newer technological frontier.

Figure 3 explores the prevalence of AI-related and computer skills across postings. This analysis identifies AI-related skills based on the methodology developed by Acemoglu et al. (2022), while broader computer skills borrow the taxonomy proposed by Deming and Kahn (2018). The left panel reports the share of postings requiring AI or computer skills in manufacturing, while the right panel broadens the analysis to the overall labor market.

Three patterns emerge from these trends. First, computer skills are substantially more prevalent in manufacturing (35 percent of postings, on average) than across all sectors (24 percent), reflecting the sector's longstanding integration of computer-based technologies, though computer requirements in both groups have declined modestly in recent months. Second, AI skill requirements show a more recent and rapid emergence: after remaining flat and modest through early 2025, AI-related requirements surged in the second half of last year, reaching 11 percent in manufacturing versus 8 percent economy-wide. These levels represent substantial growth from 2018, when Acemoglu et al. (2022) found that approximately 1 percent of vacancies—in both manufacturing and the overall economy—required AI skills. Third, manufacturing's higher AI skill requirements relative to the broader market—despite being typically classified as relatively less AI exposed—suggest that manufacturers are adopting these technologies more aggressively than conventional exposure measures might predict.

## Figure 3. Share of Postings Requiring AI and Computer Skills

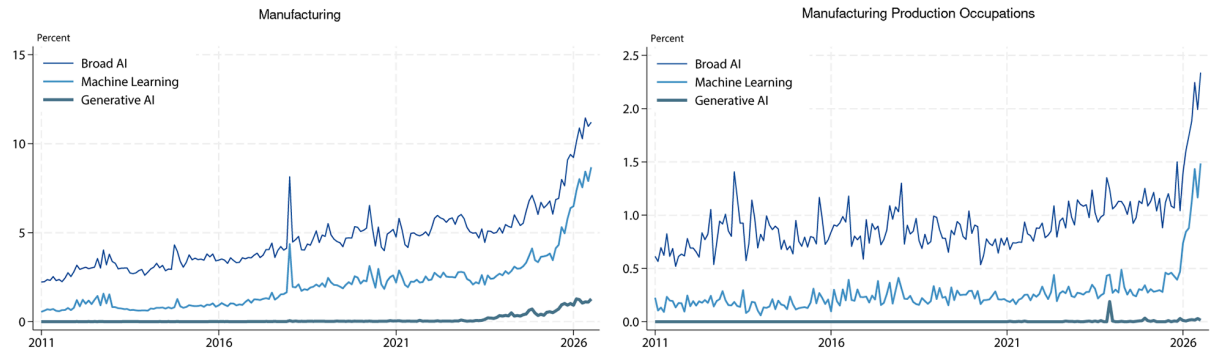



Note: Share of total new postings with AI skills or general computer skills. The left panel reflects postings for the manufacturing sector, while the right panel is for all postings. Data through July 2026.
Source: Lightcast, Job Vacancy Data.
Accessible version

Figure 4 decomposes the broad AI skills from Figure 3 into two more specific categories: machine learning and generative AI. The left panel shows manufacturing overall, while the right panel narrows in on production occupations—the sector's core workforce, representing around 50 percent of employment according to the BLS Occupational Employment and Wage Statistics, and among the least AI-exposed roles. Note the difference in axis scale between the left and right panels.

The data reveal three key patterns. First, machine learning requirements have increased notably since mid-2025, mirroring the broad AI skills patterns. Second, generative AI skills remain rare overall (under 1 percent of postings), though they have inched up over the past year. Third, production workers show the same upward trends for broad AI and machine learning but at substantially lower levels, with generative AI skills essentially absent from production postings through the first half of this year.

## Figure 4. Types of AI Skills Required in Manufacturing Postings



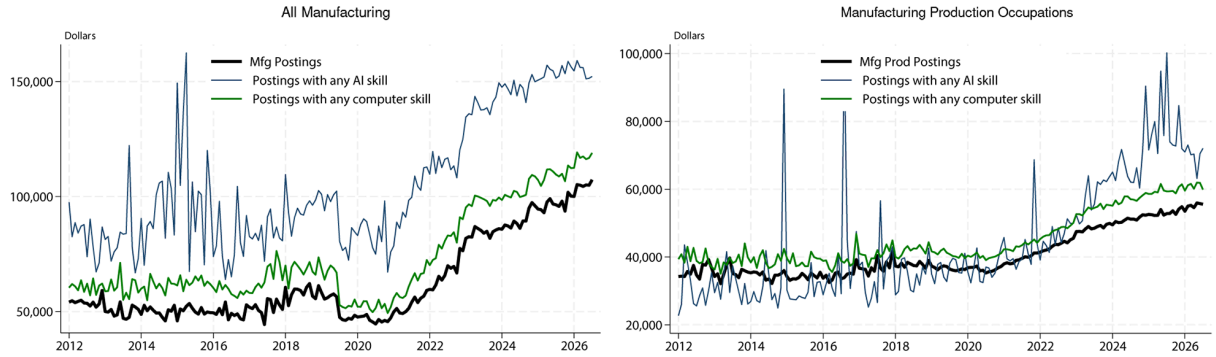

Note: Share of new postings with AI skills. The left panel refers to postings for the entire manufacturing sector, while the right panel is for production occupations (SOC code 51-0000, which includes assemblers, machinists, welders, and other roles directly involved in manufacturing products) within the manufacturing sector. Data through July 2026.
Source: Lightcast, Job Vacancy Data.
Accessible version

The evolution of advertised wages provides additional insight into how employers value these emerging skills. Figure 5 compares average posted wages for positions requiring computer skills or

AI capabilities with wages for all manufacturing postings.[3] Across all manufacturing occupations (left panel), postings requiring either skill set advertise higher wages. Computer skill wage differentials—that is, the difference in wages between postings requiring computer skills and all postings—are relatively stable across all manufacturing occupations as well as for production occupations specifically (right panel), consistent with the role of information technologies as a mature component of manufacturing work.[4]

AI skill wage differentials, however, exhibit different patterns. Among all manufacturing positions, postings requiring AI skills are associated with wages that are, on average, around 70 percent higher than all manufacturing posted wages over the sample period, with substantial volatility likely stemming from the small sample of AI-related postings and shifts in which types of positions require these skills. Interestingly, the AI wage differential emerged before the rapid increase in AI-related postings. Production occupations show a more recent shift: AI-related postings for manufacturing production workers initially displayed little or no wage differential, but the wage gap widened beginning in 2023 and has averaged roughly 30 percent since then. This pattern suggests that manufacturers may be increasingly willing to pay for AI-related capabilities even among production workers, occupations that are generally considered less exposed to AI according to task-based measures.

## Figure 5. Average Posted Wages by Skill Type




Note: Average posted wages by skill requirement. The left panel refers to postings for the entire manufacturing sector, while the right panel is for production occupations (SOC 51-0000) within the manufacturing sector. Data through July 2026.
Source: Lightcast, Job Vacancy Data.
Accessible version

These findings come with some caveats. Online job postings capture employers' stated skill requirements rather than the skills of workers ultimately hired; in addition, Lightcast data reflect positions advertised online, potentially missing hiring conducted through other channels or internal promotions. Despite these limitations, the patterns documented here suggest that manufacturers are increasingly seeking workers with AI-related skills—particularly machine learning capabilities—and are willing to offer substantial wage premia for these competencies. More broadly, hiring-based measures provide an important complement to existing approaches for studying the labor market effects of AI because they reveal how firms are adjusting their skill requirements at an early stage of technological adoption. As AI adoption continues to spread across establishments, tracking changes in advertised skill requirements and associated wage premia will be an important margin for understanding how technology affects labor demand, occupational structure, and wages in manufacturing and beyond.

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1. I would like to thank John Coglianese and Leland Crane for helpful discussions. The views expressed in the article are those of the author and do not necessarily reflect those of the Federal Reserve Board, the Federal Reserve System, or its staff. Return to text

2. The manufacturing's share of hiring demand is somewhat below its share of total employment, at 10 percent, likely reflecting lower turnover and hiring intensity in the manufacturing sector compared with other industries. Return to text

3. Wage information is not reported in all job postings. The share of postings with non-missing wage data was approximately 10 percent before the pandemic but has risen to around 50 percent in recent years. Return to text

4. As calculated, wage differentials may also reflect differences in the types of positions, locations, and employers that require these skills. Return to text

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